Yantravedh 2026

TITLE PAGE

RoadGuard AI

ML-Based Road Damage Detection Using PMPML Buses

- **Problem Statement Title:** RoadGuard AI: ML-Based Road Damage Detection and Classification System
- **Theme:** Artificial Intelligence (AI), Smart Mobility and Road Safety
- **Team Name:** **404: Ground Not Found**
- **College:** PICT (Pune Institute of Computer Technology)
- **Department:** ENTC (Electronics & Telecommunication)

Track: Hardware Hackathon | Platform: Edge-AI & PMPML Transit Fleet



PMPML Transit Bus • Real-Time AI Road Vibration Telemetry Scan (Pune)
Passive Mobile Road Inspection via Operational Public Transportation Fleet

IDEA TITLE

Proposed Solution (Describe your Idea/Solution/Prototype)

Idea Title: RoadGuard AI – Intelligent Road-Vibration Classification System

1. THE CHALLENGE (PROBLEM)

- Potholes cause accidents & vehicle damage
- Manual road surveys are slow & expensive
- Speed breakers cause false pothole alarms

SOLVED BY

2. ROADGUARD AI (SOLUTION)

- Mount MPU6050 sensor on PMPML transit buses
- Capture 6-DOF acceleration & gyro telemetry
- ML classifier categorizes road event types

DELIVERS

3. CORE INNOVATION (UNIQUENESS)

- Public buses act as autonomous crowd-sensors
- Classifies waveform patterns vs raw thresholds
- Highly scalable, sub-$10 low-cost hardware

POTHOLE (Structural Cavity)



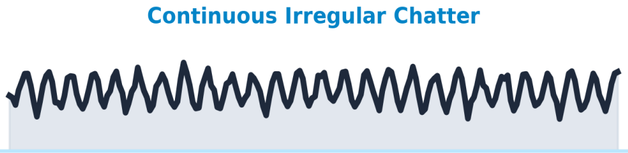
- Physical Geometry:
  Sudden depression / missing chunk of asphalt road
- Sensor Impact Signature (Z-Axis):
  Sharp downward drop (-0.5g) followed by violent rebound shock (> +3.0g). Transient duration: 150-250ms
- ML Classification Decision:
  High peak-to-peak delta + short duration
  → FLAGGED FOR MUNICIPAL ROAD REPAIR

SPEED BREAKER (Engineered Hump)



- Physical Geometry:
  Engineered rounded asphalt / rubber elevation
- Sensor Impact Signature (Z-Axis):
  Smooth, low-frequency sinusoidal rise & dip (0.8-1.5 Hz). Sustained duration: 600-900ms
- ML Classification Decision:
  Smooth symmetric profile + extended duration
  → NORMAL TRAFFIC CALMING (No Damage Alert)

ROUGH ROAD (Surface Texture)



- Physical Geometry:
  Worn aggregate, patchy asphalt tar, micro-cracks
- Sensor Impact Signature (Z-Axis):
  Persistent stochastic multi-frequency vibration without solitary extreme spikes. RMS ~0.35g
- ML Classification Decision:
  High variance baseline without isolated impulse
  → ROAD ROUGHNESS LOG (Prevents False Alarms)

TECHNICAL APPROACH

1. BUS TRAVELS
PMPML transit bus moves on regular route

2. SENSOR ACQUISITION
MPU6050 records 6-DOF accel (g) & gyro (deg/s)

3. MICROCONTROLLER
ESP32 / Arduino samples raw I2C signals @ 50-100Hz

4. SIGNAL PREPROCESSING
Bandpass filter, windowing & feature extraction (peak, RMS)

5. TRAINED ML MODEL
Random Forest classifies extracted feature vector

6. 3-CLASS DECISION
Pothole vs Speed Breaker vs Rough Road

7. LOG & MAPPING
Pothole timestamped, logged & optional GPS mapped

EXPECTED PROTOTYPE SETUP (HARDWARE)

SPACE RESERVED FOR PROTOTYPE PHOTO

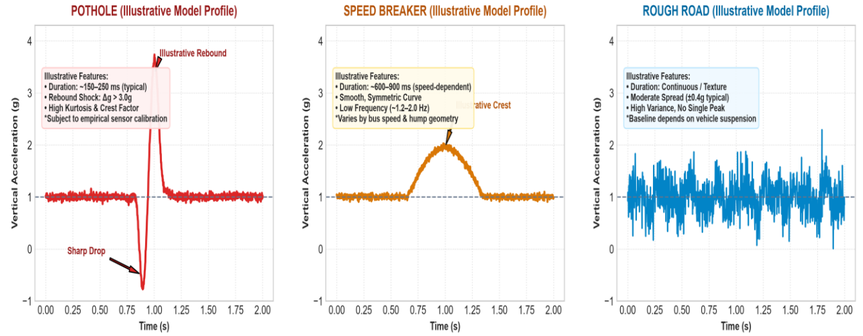
[Right-Click Image > 'Change Picture' or Paste Photo Here]

Recommended: Physical photo of MPU-6050 + ESP32 / Arduino breadboard, enclosure casing, or vehicle chassis test setup

MPU-6050 6-Axis IMU | ESP32 / Arduino MCU | Vibration-Damped Mount | Power Unit

- **Sensor:** MPU6050 6-Axis IMU (Acc ±2g to ±16g, 3-Axis Gyro)
- **Microcontroller:** ESP32 / Arduino Uno (I2C Bus @ 50–100Hz)
- **Enclosure:** Rigid IP65 Casing with Vibration-Damped Chassis Mount
- **Telemetry & Power:** 5V/3.3V Regulated; Optional NEO-6M GPS & GSM
- **Firmware Stack:** Arduino C++, Scikit-learn Random Forest Model

VIBRATION SIGNATURE ANALYSIS (Z-AXIS ACCEL)



*Note: Waveforms and threshold values are illustrative model profiles from road-vibration literature; exact operational values will be empirically calibrated using onboard bus sensor data.

ILLUSTRATIVE WAVEFORM DISCRIMINATION PATTERNS

- **Pothole (Cavity):** Sharp negative drop followed by high rebound spike (illustrative ~150–250ms transient). Characterized by high kurtosis and high delta-g relative to baseline.
- **Speed Breaker (Hump):** Smooth, symmetrical low-frequency elevation (illustrative ~600–900ms duration at ~20–30 km/h). Distinct low-frequency sinusoidal profile without sharp shock.
- **Rough Road (Texture):** Continuous irregular vibration chatter (illustrative ±0.4g spread around 1.0g gravity baseline). High variance without isolated high-energy peaks.
- ***Calibration Note:** Waveforms and thresholds shown are illustrative benchmarks from published research; final operational thresholds will be empirically calibrated using onboard bus sensor data.*

5-STEP ML WORKFLOW

1. DATA ACQUISITION
Collect 6-DOF accel & gyro streams on routes

2. PREPROCESSING
Zero-drift calibration & gravity removal (1g)

3. FEATURE EXTRACTION
Peak-to-peak delta, RMS, duration, kurtosis

4. MODEL TRAINING
Random Forest classifier cross-validation

5. REAL-TIME INFERENCE
Categorize: Pothole / Speed Breaker / Rough

Prototype Configuration Note:
Exact ML algorithm, deployment platform (edge MCU vs companion PC), and optional modules depend on final prototype setup.

FEASIBILITY AND VIABILITY



1. FEASIBILITY (Why It Works)

[✓] Sub-$2 Sensor Hardware: Uses commodity MPU6050 MEMS sensor (sub-$2), keeping build cost negligible.

[✓] Rapid Microcontroller Setup: Easily prototyped with ESP32 / Arduino Uno interfaced with a laptop for data capture.

[✓] Lightweight Machine Learning: Random Forest runs with tiny compute footprint—zero costly GPU infrastructure needed.

[✓] Proposed Field Testing: Planned experimental trials and calibration across diverse Pune road surfaces and PMPML transit routes using labeled sensor runs.

[✓] Modular Architecture: Decoupled data collection, preprocessing, and classification for easy incremental upgrades.

2. POTENTIAL CHALLENGES

[!] Bus Chassis & Suspension Variance: Different bus models and mounting locations produce shifted baseline vibration readings.

[!] Vibration Waveform Overlap: Steep speed breakers crossed at speed can mimic acceleration peaks of shallow potholes.

[!] Vehicle Speed & Payload Dynamics: Fluctuating passenger weight and driving speeds influence raw vibration amplitudes.

[!] Representative Dataset Needs: Accurate ML categorization requires high-quality, ground-truthed training datasets.

[!] Transit Environmental Shocks: Hardware deployed on city buses faces continuous vibration, dust, and temperature shifts.

3. MITIGATION STRATEGIES

[★] Standardized Rigid Axle Mount: Enclose sensor in rigid, damped IP65 casing bolted to the chassis to ensure clean vibration transfer.

[★] Multi-Speed Calibration Data: Collect labeled calibration data across varying speeds (15–50 km/h) and varied pavement textures.

[★] Multi-Feature Pattern Decision: Classify complete waveforms (duration, RMS, kurtosis, peak-to-peak delta) rather than simple thresholds.

[★] Rigorous Cross-Validation: Evaluate model on separate, unseen test routes with k-fold validation before operational claims.

[★] Adaptive Baseline Filtering: Implement dynamic moving-average filtering to compensate for bus payload changes.

Feasibility Summary: Sub-$15 Hardware BOM • High Municipal Scalability • Robust Multi-Feature ML Classification

IMPACT AND BENEFITS

SOCIAL IMPACT (ROAD SAFETY & COMMUTER WELFARE)

- **Safer Urban Commutes:** Continuous passive road auditing enables proactive maintenance for safer roads.
- **Prevents Fatal Accidents:** Early detection of hazardous potholes directly prevents two-wheeler skids & vehicle damage.

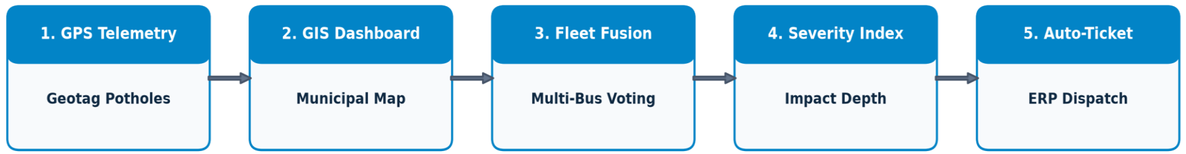
ECONOMIC IMPACT (COST EFFICIENCY & RESOURCE ALLOCATION)

- **Reduces Survey Expenditure:** Automates audits, drastically cutting manual survey labor & dedicated vehicle fuel costs.
- **Data-Driven Prioritization:** Provides objective damage records to help municipal authorities triage repair budgets.

ENVIRONMENTAL & OPERATIONAL IMPACT (SUSTAINABLE MOBILITY)

- **Zero Dedicated Survey Emissions:** Uses scheduled in-service buses—zero extra survey trips or added carbon emissions.
- **Non-Intrusive City Coverage:** Runs silently during regular passenger service without traffic disruptions or road closures.

FUTURE SCOPE & SYSTEM ROADMAP (PROPOSED EXTENSIONS)



- **Proposed Stages: [1] GPS Geotagging → [2] GIS Municipal Map → [3] Multi-Bus Fleet Fusion → [4] Severity Estimation → [5] Municipal ERP Ticketing.**
- Objective: Enable automated work order generation directly into municipal maintenance workflows.



PROPOSED MUNICIPAL GIS CONCEPT (ILLUSTRATIVE DEMO DASHBOARD)

Smart-City Fleet Monitoring Concept (Pune Municipal Corporation):

- Illustrative Demo Data Notice: All statistics, report counts (e.g., 128 reports, 42 active buses), incident alerts, and map pin coordinates shown on this interface are synthetic demonstration data for UI/concept visualization only.
- Prototype Scope Notice: Real-time GPS mapping, automated cloud telemetry, and centralized municipal dashboard integration represent proposed future extensions and are not part of the initial standalone hardware prototype.
- Note: No guaranteed financial savings, confirmed government integration, or unverified detection accuracies are claimed.

RESEARCH AND REFERENCES



1. HARDWARE SPECIFICATION (MPU-6050)

- **Primary Source:** TDK InvenSense MPU-6050 Specification (Doc PS-MPU-6000A-00)
- **Technical Scope:** Integrated 6-axis MEMS MotionTracking (±2g to ±16g Accel, 3-Axis Gyro, I2C interface).
- **Project Role:** High-rate road vibration capture and transient chassis impact sampling.

2. MACHINE LEARNING FRAMEWORK (RANDOM FOREST)

- **Primary Source:** Pedregosa et al., "Scikit-learn: Machine Learning in Python" (JMLR 2011)
- **Technical Scope:** Supervised ensemble classification using RandomForestClassifier algorithm.
- **Project Role:** Multi-class categorization (Pothole vs Speed Breaker vs Rough Road) with noise robustness.

3. VALIDATION METHODOLOGY (EVALUATION PROTOCOLS)

- **Primary Source:** Buitinck et al., "API Design for Machine Learning Software" (ECML PKDD)
- **Technical Scope:** Train/test split protocols (train_test_split), k-fold cross-validation, and multi-class metrics.
- **Project Role:** Ensures model generalization across independent test routes; prevents test data leakage.

4. PEER-REVIEWED RESEARCH (TRANSIT ROAD SENSING)

- **Mednis et al. (IEEE DCOSS 2011):** Real-time pothole detection using mobile accelerometers.
- **De Zoysa et al. (IEEE ICIAfS 2007):** Public transport bus-based road condition monitoring.
- **Mohan et al. (ACM SenSys 2008):** Nericell: Mobile sensing for road and traffic classification.

PROJECT SUMMARY & CORE INNOVATION

"RoadGuard AI combines low-cost vibration sensing and Machine Learning to distinguish potholes from common road irregularities, supporting smarter road inspection."

Hardware Hackathon Track: AI, Smart Mobility and Road Safety | Target Deployment: Public Transit Bus Fleets (PMPML, Pune)